A

LogReg

Train country \ Test country	Armenia 2015-2016	Bolivia 2008	Cameroon 2018	Congo DRC 2021–2022	Gabon 2019–2021	Gambia 2019–2020	Ghana 2022	Guatemala 2014–2015	Haiti 2016-2017	Jordan 2023	Kyrgyz Republic 2012	Liberia 2019–2020	Malawi 2024	Mali 2023–2024	Rwanda 2019–2020	Sierra Leone 2019
Armenia 2015-2016		0.66	0.52	0.50	0.47	0.55	0.57	0.56	0.54	0.53	0.54	0.52	0.56	0.56	0.54	0.55
Bolivia 2008	0.68		0.61	0.59	0.60	0.63	0.54	0.60	0.56	0.54	0.62	0.56	0.63	0.63	0.61	0.60
Cameroon 2018	0.59	0.53		0.62	0.63	0.69	0.68	0.66	0.62	0.63	0.60	0.56	0.64	0.65	0.62	0.62
Congo DRC 2021–2022	0.58	0.51	0.64		0.63	0.68	0.66	0.64	0.63	0.60	0.59	0.57	0.65	0.65	0.61	0.62
Gabon 2019–2021	0.52	0.46	0.62	0.60		0.66	0.65	0.60	0.61	0.60	0.56	0.56	0.63	0.63	0.59	0.60
Gambia 2019–2020	0.49	0.43	0.63	0.61	0.63		0.67	0.60	0.62	0.64	0.53	0.57	0.61	0.65	0.57	0.62
Ghana 2022	0.55	0.46	0.64	0.62	0.63	0.70		0.63	0.63	0.63	0.57	0.58	0.64	0.66	0.61	0.62
Guatemala 2014–2015	0.68	0.66	0.64	0.61	0.59	0.65	0.66		0.64	0.62	0.64	0.60	0.67	0.66	0.65	0.62
Haiti 2016-2017	0.60	0.49	0.63	0.60	0.61	0.67	0.67	0.67		0.62	0.60	0.59	0.66	0.66	0.62	0.61
Jordan 2023	0.56	0.47	0.63	0.61	0.63	0.69	0.68	0.63	0.63		0.58	0.58	0.63	0.65	0.59	0.63
Kyrgyz Republic 2012	0.70	0.72	0.60	0.57	0.59	0.63	0.60	0.66	0.63	0.59		0.59	0.65	0.65	0.65	0.60
Liberia 2019–2020	0.41	0.40	0.57	0.57	0.59	0.61	0.59	0.55	0.59	0.56	0.51		0.59	0.60	0.53	0.54
Malawi 2024	0.64	0.55	0.63	0.61	0.60	0.65	0.66	0.69	0.66	0.63	0.63	0.61		0.66	0.64	0.62
Mali 2023–2024	0.59	0.50	0.63	0.61	0.63	0.69	0.68	0.66	0.65	0.62	0.59	0.58	0.66		0.62	0.63
Rwanda 2019–2020	0.72	0.72	0.62	0.59	0.60	0.64	0.64	0.66	0.62	0.60	0.65	0.59	0.65	0.65		0.62
Sierra Leone 2019	0.63	0.70	0.62	0.59	0.61	0.68	0.66	0.62	0.59	0.57	0.60	0.54	0.62	0.64	0.61	

B

LightGBM

Train country \ Test country	Armenia 2015-2016	Bolivia 2008	Cameroon 2018	Congo DRC 2021–2022	Gabon 2019–2021	Gambia 2019–2020	Ghana 2022	Guatemala 2014–2015	Haiti 2016-2017	Jordan 2023	Kyrgyz Republic 2012	Liberia 2019–2020	Malawi 2024	Mali 2023–2024	Rwanda 2019–2020	Sierra Leone 2019
Armenia 2015-2016		0.66	0.54	0.51	0.52	0.52	0.57	0.59	0.58	0.56	0.59	0.57	0.56	0.55	0.57	0.54
Bolivia 2008	0.66		0.63	0.59	0.60	0.63	0.63	0.67	0.62	0.60	0.62	0.56	0.63	0.62	0.61	0.60
Cameroon 2018	0.67	0.66		0.60	0.60	0.63	0.63	0.67	0.61	0.60	0.60	0.56	0.63	0.62	0.63	0.60
Congo DRC 2021–2022	0.63	0.66	0.64		0.61	0.64	0.65	0.68	0.58	0.59	0.61	0.57	0.63	0.60	0.63	0.60
Gabon 2019–2021	0.56	0.58	0.62	0.60		0.63	0.65	0.62	0.59	0.60	0.59	0.57	0.60	0.62	0.60	0.61
Gambia 2019–2020	0.61	0.63	0.63	0.59	0.60		0.65	0.64	0.63	0.61	0.63	0.59	0.63	0.65	0.61	0.61
Ghana 2022	0.50	0.50	0.60	0.55	0.60	0.65		0.55	0.61	0.58	0.52	0.59	0.55	0.64	0.54	0.61
Guatemala 2014–2015	0.65	0.66	0.63	0.61	0.59	0.65	0.65		0.62	0.60	0.63	0.57	0.68	0.62	0.64	0.62
Haiti 2016-2017	0.65	0.63	0.59	0.58	0.57	0.65	0.65	0.66		0.60	0.62	0.60	0.65	0.66	0.59	0.61
Jordan 2023	0.52	0.55	0.59	0.55	0.60	0.66	0.61	0.59	0.61		0.56	0.56	0.58	0.61	0.55	0.58
Kyrgyz Republic 2012	0.67	0.66	0.55	0.55	0.56	0.59	0.60	0.63	0.62	0.58		0.59	0.63	0.63	0.62	0.56
Liberia 2019–2020	0.60	0.53	0.55	0.54	0.55	0.56	0.63	0.60	0.61	0.57	0.59		0.63	0.60	0.58	0.53
Malawi 2024	0.56	0.56	0.61	0.59	0.58	0.63	0.64	0.67	0.62	0.56	0.59	0.59		0.62	0.62	0.60
Mali 2023–2024	0.59	0.62	0.59	0.57	0.59	0.66	0.61	0.60	0.62	0.59	0.59	0.59	0.59		0.55	0.59
Rwanda 2019–2020	0.63	0.64	0.61	0.59	0.56	0.60	0.65	0.64	0.59	0.54	0.60	0.56	0.63	0.61		0.59
Sierra Leone 2019	0.63	0.63	0.59	0.60	0.59	0.55	0.60	0.62	0.61	0.56	0.63	0.55	0.63	0.58	0.63	

C

XGBoost

Train country \ Test country	Armenia 2015-2016	Bolivia 2008	Cameroon 2018	Congo DRC 2021–2022	Gabon 2019–2021	Gambia 2019–2020	Ghana 2022	Guatemala 2014–2015	Haiti 2016-2017	Jordan 2023	Kyrgyz Republic 2012	Liberia 2019–2020	Malawi 2024	Mali 2023–2024	Rwanda 2019–2020	Sierra Leone 2019
Armenia 2015-2016		0.66	0.53	0.49	0.54	0.53	0.57	0.56	0.56	0.56	0.60	0.56	0.54	0.52	0.58	0.54
Bolivia 2008	0.66		0.63	0.59	0.61	0.63	0.64	0.68	0.63	0.61	0.64	0.57	0.65	0.63	0.62	0.60
Cameroon 2018	0.67	0.65		0.61	0.61	0.63	0.64	0.67	0.61	0.60	0.61	0.56	0.64	0.64	0.64	0.61
Congo DRC 2021–2022	0.65	0.64	0.64		0.61	0.65	0.65	0.67	0.60	0.60	0.61	0.58	0.63	0.62	0.63	0.60
Gabon 2019–2021	0.57	0.58	0.62	0.58		0.65	0.66	0.63	0.61	0.59	0.59	0.58	0.61	0.62	0.59	0.61
Gambia 2019–2020	0.61	0.62	0.63	0.59	0.61		0.66	0.63	0.63	0.62	0.63	0.59	0.63	0.65	0.61	0.61
Ghana 2022	0.55	0.53	0.60	0.55	0.62	0.67		0.57	0.61	0.57	0.50	0.60	0.55	0.64	0.54	0.61
Guatemala 2014–2015	0.65	0.68	0.64	0.61	0.59	0.66	0.66		0.63	0.60	0.62	0.58	0.67	0.63	0.64	0.61
Haiti 2016-2017	0.64	0.64	0.60	0.57	0.59	0.65	0.66	0.65		0.61	0.62	0.61	0.65	0.66	0.59	0.60
Jordan 2023	0.55	0.56	0.59	0.56	0.60	0.66	0.61	0.60	0.61		0.58	0.55	0.59	0.62	0.55	0.56
Kyrgyz Republic 2012	0.65	0.65	0.54	0.54	0.55	0.59	0.58	0.62	0.60	0.58		0.58	0.61	0.59	0.62	0.55
Liberia 2019–2020	0.62	0.57	0.56	0.56	0.56	0.59	0.62	0.62	0.62	0.58	0.61		0.63	0.61	0.59	0.54
Malawi 2024	0.62	0.60	0.62	0.60	0.59	0.64	0.63	0.68	0.63	0.58	0.62	0.59		0.63	0.62	0.60
Mali 2023–2024	0.58	0.64	0.61	0.58	0.61	0.68	0.63	0.64	0.64	0.62	0.60	0.60	0.61		0.56	0.61
Rwanda 2019–2020	0.64	0.64	0.62	0.59	0.57	0.61	0.64	0.64	0.59	0.56	0.62	0.56	0.63	0.60		0.59
Sierra Leone 2019	0.62	0.64	0.60	0.59	0.60	0.60	0.61	0.62	0.62	0.57	0.63	0.55	0.63	0.59	0.61	

D

TabPFN

Train country \ Test country	Armenia 2015-2016	Bolivia 2008	Cameroon 2018	Congo DRC 2021–2022	Gabon 2019–2021	Gambia 2019–2020	Ghana 2022	Guatemala 2014–2015	Haiti 2016-2017	Jordan 2023	Kyrgyz Republic 2012	Liberia 2019–2020	Malawi 2024	Mali 2023–2024	Rwanda 2019–2020	Sierra Leone 2019
Armenia 2015-2016		0.68	0.58	0.53	0.57	0.61	0.62	0.62	0.63	0.58	0.64	0.59	0.62	0.64	0.64	0.59
Bolivia 2008	0.69		0.64	0.60	0.61	0.67	0.67	0.70	0.64	0.62	0.66	0.59	0.67	0.67	0.65	0.62
Cameroon 2018	0.66	0.66		0.63	0.63	0.67	0.68	0.70	0.64	0.63	0.64	0.59	0.66	0.66	0.66	0.63
Congo DRC 2021–2022	0.61	0.62	0.65		0.62	0.66	0.68	0.70	0.63	0.62	0.61	0.60	0.66	0.66	0.65	0.63
Gabon 2019–2021	0.60	0.61	0.64	0.61		0.66	0.68	0.65	0.62	0.62	0.61	0.59	0.63	0.65	0.61	0.63
Gambia 2019–2020	0.61	0.65	0.63	0.59	0.63		0.67	0.65	0.62	0.63	0.63	0.58	0.63	0.66	0.62	0.62
Ghana 2022	0.65	0.66	0.65	0.61	0.64	0.69		0.68	0.64	0.63	0.65	0.61	0.65	0.67	0.63	0.63
Guatemala 2014–2015	0.68	0.68	0.64	0.62	0.61	0.67	0.67		0.65	0.62	0.63	0.60	0.69	0.66	0.65	0.63
Haiti 2016-2017	0.68	0.66	0.63	0.60	0.60	0.67	0.67	0.70		0.64	0.65	0.62	0.67	0.67	0.63	0.62
Jordan 2023	0.60	0.59	0.62	0.59	0.62	0.69	0.66	0.66	0.65		0.62	0.60	0.63	0.67	0.61	0.60
Kyrgyz Republic 2012	0.72	0.69	0.58	0.56	0.57	0.61	0.61	0.67	0.64	0.60		0.60	0.65	0.65	0.64	0.59
Liberia 2019–2020	0.66	0.63	0.60	0.59	0.59	0.64	0.65	0.68	0.65	0.62	0.65		0.67	0.67	0.63	0.58
Malawi 2024	0.64	0.63	0.63	0.61	0.60	0.64	0.65	0.72	0.66	0.61	0.63	0.62		0.65	0.65	0.61
Mali 2023–2024	0.66	0.68	0.63	0.59	0.63	0.69	0.67	0.68	0.66	0.63	0.63	0.60	0.66		0.61	0.62
Rwanda 2019–2020	0.69	0.67	0.63	0.60	0.60	0.62	0.65	0.67	0.63	0.60	0.65	0.58	0.65	0.63		0.61
Sierra Leone 2019	0.65	0.67	0.64	0.61	0.63	0.68	0.67	0.68	0.62	0.61	0.63	0.59	0.65	0.66	0.64	

Test country

Train country

AUC–ROC

0.70

0.65

0.60

0.55

0.50

0.45

0.40